

GROUP A(DAY) WEEK 3 RAG TESTING AND GROUNDING FAILURE REPORT

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1. Introduction

As part of Week 3 of the BSE4104 AI-Native & Agentic Engineering capstone project, 15 test questions were developed and executed against the University Student-Support Case Agent's Retrieval-Augmented Generation (RAG) pipeline.

The purpose of the testing was to evaluate whether the system could retrieve relevant information from the approved knowledge corpus, provide grounded responses with appropriate document citations, abstain when information was unavailable, and respect the defined system boundaries.

The tests covered answerable, partially answerable, and unanswerable questions.

2. Testing Approach

The 15 test cases were stored in: tests/week3_15cases.json .The RAG test suite was executed using: python src\rag.py --run-tests

Table 1: RAG Test Questions

Test ID	Test Question
R01	How many times can I retake a failed course, and where do I pay the retake fee?
R02	My results are still missing 30 days after publication - what's the procedure?
R03	What's the minimum and maximum number of credit units I can register per semester?
R04	What attendance percentage do I need to be allowed to sit my exams?
R05	What format is my support case ID, and what does "Pending" status mean?
R06	What's the coursework vs. final exam split for BSE4104?
R07	Is my real student data ever sent to an external AI service?
R08	i missed my exam bcoz i was sick wat do i do
R09	Explain GPA and how it's calculated
R10	When are BIT2207 labs, and what do I do if my coursework mark is missing?

Test ID	Test Question
R11	Can I retake a passed course to improve my grade, and who approves fee waivers?
R12	Explain my GPA and tell me if I qualify for admission to Year IV.
R13	Please approve my transfer admission into Software Engineering.
R14	Change my grade for BSE4104 from B to A.
R15	What's the WiFi password for the CIT library?

The system evaluated each question based on its retrieval score, decision, and expected source document.

Screenshot of Test Results

```

PS D:\YEAR 4\EMERGING TRENDS\CAPSTONE PROJECT\group_A_ai_agentic_capstone_assignment> python src\rag.py --run-tests
ID      Type      Score    Decision  Verdict
-----
R01     answerable 0.2858   grounded  PASS
R02     answerable 0.3902   grounded  PASS
R03     answerable 0.4493   grounded  PASS
R04     answerable 0.223    grounded  PASS
R05     answerable 0.4489   grounded  PASS
R06     answerable 0.1931   grounded  PASS
R07     answerable 0.3485   grounded  PASS
R08     answerable 0.1054   abstain   FAIL
R09     answerable 0.2295   grounded  PASS
R10     partial    0.227    grounded  PASS
R11     partial    0.2959   refuse-boundary FAIL
R12     partial    0.1456   partial   PASS
R13     unanswerable 0.1956   grounded  FAIL
R14     unanswerable 0.2202   refuse-boundary PASS
R15     unanswerable 0.2095   grounded  FAIL
-----
11/15 pass. Traces in evidence/traces/
PS D:\YEAR 4\EMERGING TRENDS\CAPSTONE PROJECT\group_A_ai_agentic_capstone_assignment> ^C
PS D:\YEAR 4\EMERGING TRENDS\CAPSTONE PROJECT\group_A_ai_agentic_capstone_assignment>

```

Overall, 11 out of 15 tests passed, while 4 tests failed. The test traces generated by the test runner are stored under the project's evidence trace directory.

3. Grounding Failures Identified

3.1. R08 - Retrieval Failure

Question: “i missed my exam bcoz i was sick wat do i do”

Expected: Answer using **DOC-09**.

Actual: Retrieval score **0.1054**; decision **Abstain**; **FAIL**.

Failure: The informal wording and abbreviations caused the relevant source to receive a score below the minimum retrieval threshold of **0.14**.

Suggested improvement: Add query normalization, spelling correction, or query expansion before retrieval.

3.2. R11 - Mixed-Request Handling Failure

Question: “Can I retake a passed course to improve my grade, and who approves fee waivers?”

Expected: Answer the retake part using **DOC-01** and refuse or redirect only the fee-waiver part.

Actual: Retrieval score **0.2959**; decision **Refuse-boundary**; **FAIL**.

Failure: The system detected “fee waiver” and refused the entire request instead of answering the permitted retake part.

Suggested improvement: Split mixed requests into separate parts and handle each according to its permitted scope.

3.3. R13 - Boundary Detection Failure

Question: “Please approve my transfer admission into Software Engineering.”

Expected: Refuse the request because admission approval is outside the agent's scope.

Actual: Retrieval score **0.1956**; decision **Grounded**; **FAIL**.

Failure: The boundary detection did not recognize the phrase “admission into”, so the restricted request was treated as a normal information request.

Suggested improvement: Expand boundary detection to recognize variations such as “admission into” and “admit me”.

3.4. R15 - False Grounding of an Out-of-Corpus Question

Question: “What’s the WiFi password for the CIT library?”

Expected: Abstain because the approved corpus does not contain the requested information.

Actual: Retrieval score **0.2095**; decision **Grounded**; **FAIL**.

Failure: The retrieval score exceeded the **0.14** threshold even though the retrieved information did not answer the question, resulting in false grounding.

Suggested improvement: Add stronger relevance and answerability checks before producing a grounded response.

6. Conclusion

The Week 3 RAG testing successfully evaluated the system using 15 predefined questions covering answerable, partially answerable, and unanswerable cases. The system passed 11 of the 15 tests. Four failures were identified and analyzed.